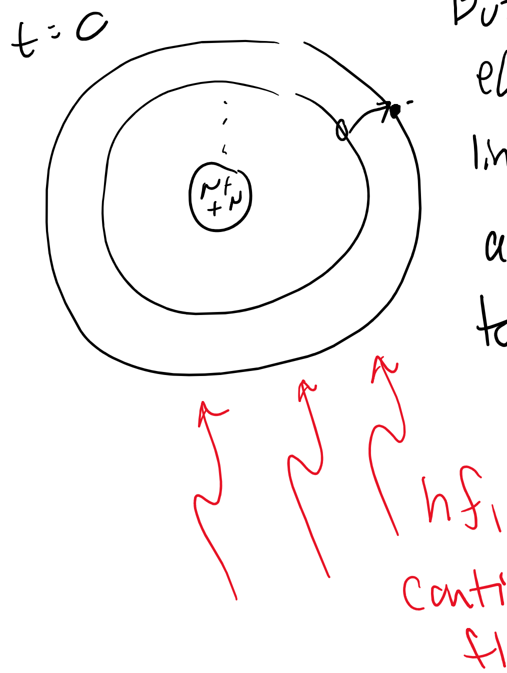
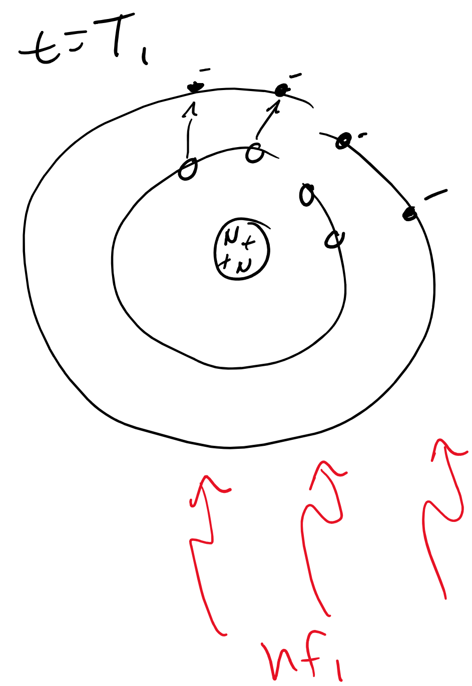
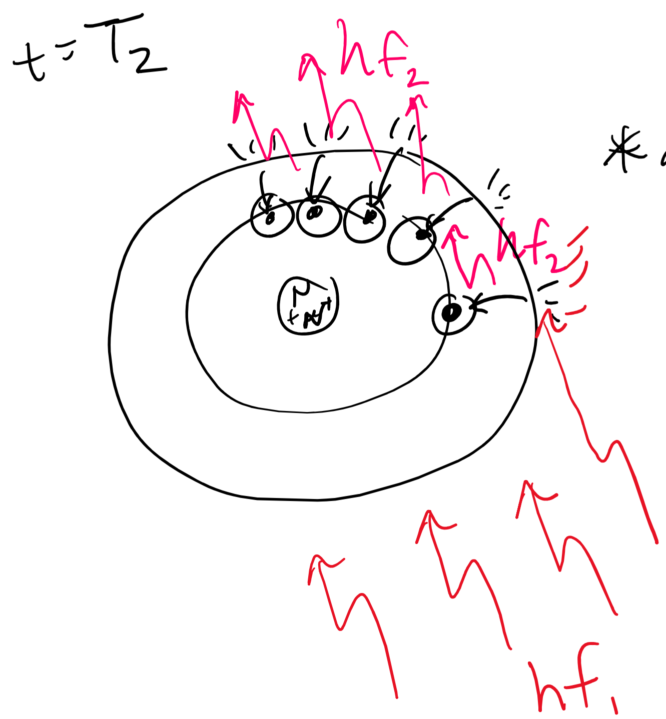
Effect 2



But the electron lingers a bit allowing more to jump to that state





* all falling together because hf_1 "hits" an electron that is "hanging out" in the excited state, causing it to "fall" creating a chain Reaction

* when this chain Reaction happens, $f_1 \approx f_2$ photon generated in the medium have approx. same phase + Polarization → coherent!

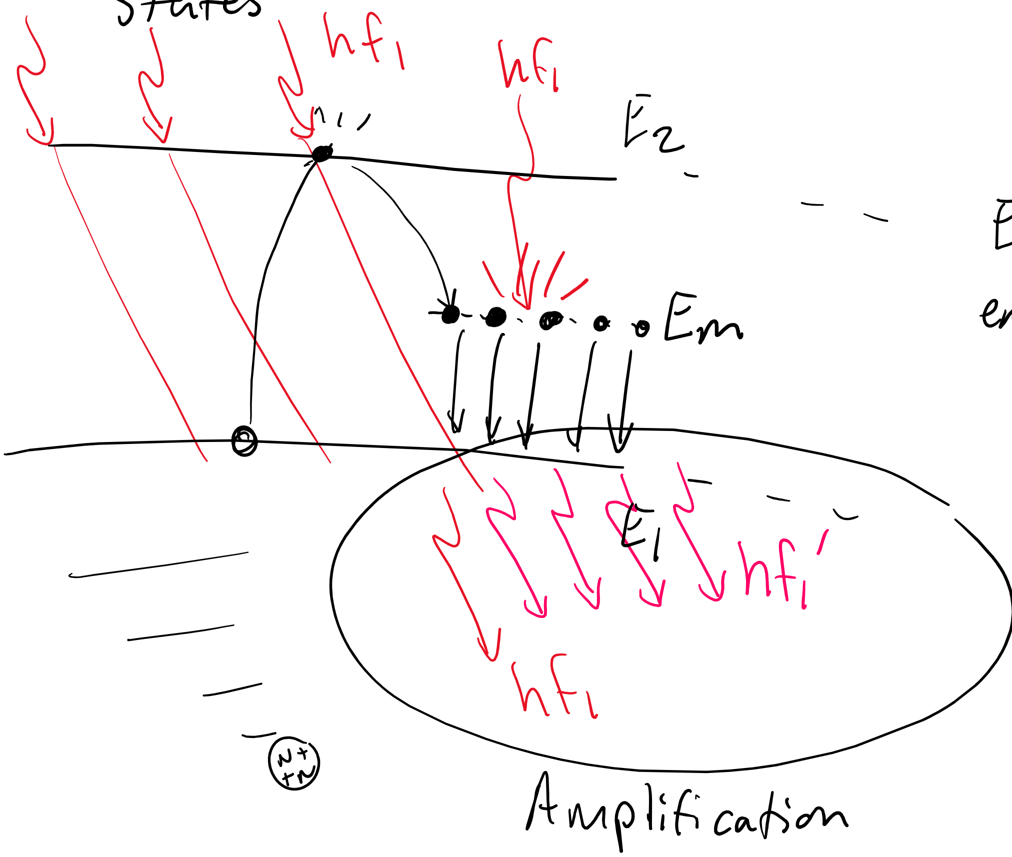
* the medium exhibits <u>optical Gain</u>.

* the act of moving electrons to a state where they "hang out" (meta-stable state) is called

$\boxed{\text{Population Inversion}}$

* 2- state system is very difficult.

* Most Population Inversions Require 3 or more states



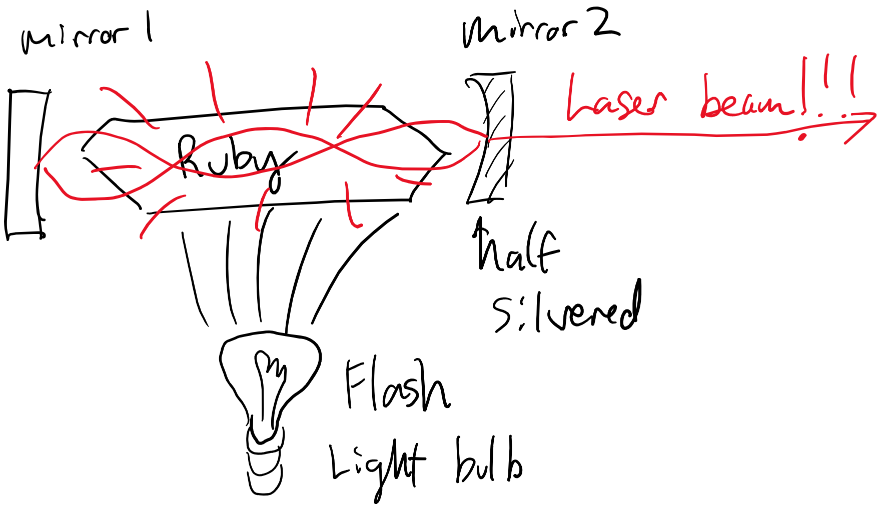
E_m is the meta-stable energy level

* the act of building up the electrons in the E_m band is called <u>Pumping</u>

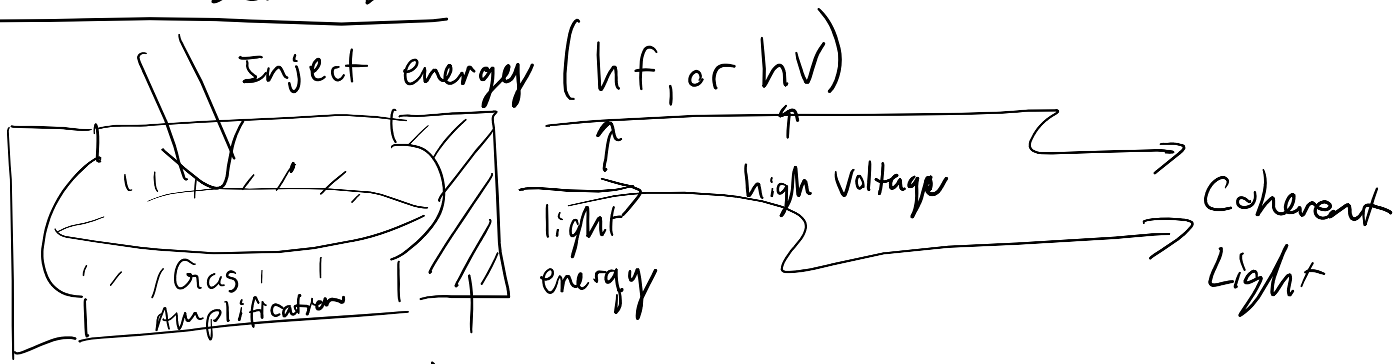
the phenomena

* Light Amplification by Stimulated Emission of Radiation (LASER)

first Laser

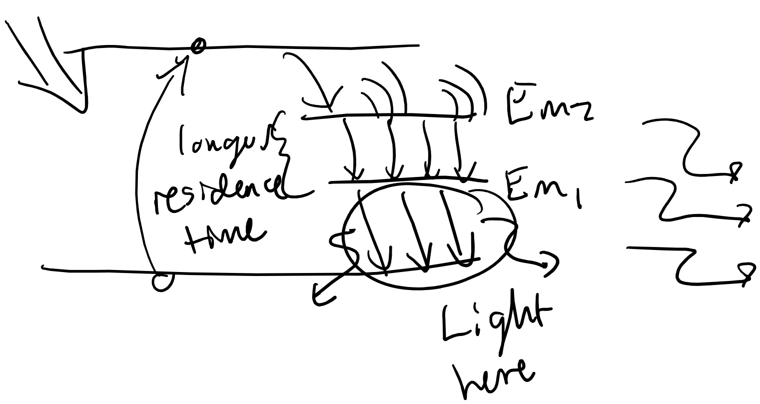


Gas Laser Era



half silverd
(or Q-switch in modern times)

* most Modern Laser are 4-energy Level process (Nd:YAG)



- gas Laser are highly temp dependent ($\frac{3}{2} k_B T$)